

Number Patterns & Sequences

Mastering Rules, Growing & Shrinking Sequences, and Finding the Nth Term

1. What is a Number Pattern?

A **number pattern** is a list of numbers that follow a specific rule. Think of it like a recipe where each step tells you exactly how to get to the next number!

A **sequence** is simply an ordered set of numbers. Each individual number in a sequence is called a **term**.

Examples of Sequences:

- 1, 3, 5, 7, 9... → Rule: Add 2 each time (+2).
- 10, 9, 8, 7, 6... → Rule: Subtract 1 each time (-1).
- 5, 10, 15, 20, 25... → Rule: Add 5 each time (+5).

2. How to Find the Term-to-Term Rule

To find the rule between consecutive numbers in a sequence, follow these 3 detective steps:

1. **Look carefully** at two numbers that are next to each other in the sequence.
2. **Determine the change:** Ask yourself, is the number getting bigger (adding/multiplying) or smaller (subtracting/dividing)? By how much?
3. **State the rule in words:** For example, "Add 3 each time" or "Subtract 5 each time."

3. Growing vs. Shrinking Patterns

• Growing Patterns (Increasing)

Growing patterns get bigger step by step. Each number is larger than the one before it.

Examples:

- 2, 4, 8, 16, 32... (Multiply by 2 each time)
- 10, 20, 30, 40... (Add 10 each time)

• Shrinking Patterns (Decreasing)

Shrinking patterns get smaller step by step. Each number is smaller than the one before it.

Examples:

- 100, 90, 80, 70... (Subtract 10 each time)
- 64, 32, 16, 8... (Divide by 2 each time)

4. Position Rules and The Nth Term

Every term in a pattern has a **position**:

- **Position 1 ($n = 1$)**: 1st number in the sequence
- **Position 2 ($n = 2$)**: 2nd number in the sequence
- **Position 3 ($n = 3$)**: 3rd number in the sequence, and so on.

What is the "Nth Term"?

The letter n represents any position number. The **nth term rule** is a mathematical formula that lets you find the exact value of ANY term in a sequence without having to write out all the numbers before it!

How to Find and Use the Nth Term Formula:

For simple linear sequences where the starting value corresponds to the step size:

1. **Step 1**: Find the common difference (the amount added each step). Let's call this d .
2. **Step 2**: Multiply the position number n by this difference: **Term** = $d \times n$.
3. **Step 3**: To find a specific term (like the 50th term), substitute $n = 50$ into your formula!

Worked Example 1: The Sequence 4, 8, 12, 16, 20...

Let's find the position rule and calculate the 10th term:

Position ()	Calculation ()	Term Value
1st ($n = 1$)	4×1	4
2nd ($n = 2$)	4×2	8
3rd ($n = 3$)	4×3	12
4th ($n = 4$)	4×4	16
10th ($n = 10$)	4×10	40

Detective Tip: Why use the Nth Term?

Imagine if your teacher asked for the **100th term** of 5, 10, 15, 20...! Adding 5 a hundred times would take forever. But using the nth term formula ($5 \times n$), you can calculate it instantly: $5 \times 100 = 500$!

Student Practice Worksheet

Topic: Number Patterns, Sequences & The Nth Term | Grade 4

Name: _____

Date: _____

Section A: Identify the Pattern & Complete the Sequence

Find the rule and fill in the missing numbers for each sequence below:

1. 3, 6, 9, __, 15, __ → Rule: _____

2. 40, 35, 30, __, 20, __ → Rule: _____

3. 6, 12, 18, 24, __, __ → Rule: _____

Section B: Growing or Shrinking?

State whether each pattern is **Growing** or **Shrinking**, then write the rule:

4. 100, 80, 60, 40, 20 → Type: _____ | Rule: _____

5. 7, 14, 21, 28, 35 → Type: _____ | Rule: _____

Section C: Using Position Rules & The Nth Term

6. For the sequence 5, 10, 15, 20, 25...

a) What is the nth term formula for this sequence? _____

b) Use your formula to find the **12th term**:

7. A sequence follows the rule: **Term** = $8 \times n$ (where **n** is the position number).

a) Find the 1st, 2nd, and 3rd terms of this sequence: ____, ____, ____

b) Find the **15th term** of this sequence:

Section D: Word Problem Challenge

8. Aarav saves money in his piggy bank every week. On Week 1 he saves ₹6, on Week 2 he saves ₹12, and on Week 3 he saves ₹18. If this pattern continues:

a) Write the rule for Aarav's weekly savings.

b) How much money will Aarav save in **Week 8**? Use the position rule (**n = 8**) to show your work.